Deterministic Evidence Verification in Parameter-Constrained Local LLMs

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Abstract?Hallucination rates in small language models (1B to 3B parameters) remain high when performing extractive scientific synthesis. We propose a deterministic evidence verification layer that breaks generated responses into atomic factual claims and validates n-gram overlap, numerical values, and bounding-box coordinates against retrieved source passages. Empirical trials show hallucinated citations drop by 78.4% without requiring expensive model fine-tuning.

1. Introduction & Methodology

When local open-weight models such as LLaMA-3.2 (1B/3B) or Qwen-2.5 (1.5B) run on consumer CPUs, their generation cannot be assumed mathematically sound. In high-stakes research intelligence, asserting an unverified claim can compromise scientific integrity.

2. Deterministic Claim Decomposition & Alignment

Our verification layer operates in three discrete stages:

1. Atomic claim decomposition: Answer strings are partitioned into verifiable propositions.
2. Lexical and numeric alignment: Exact matching of reported metric percentages (e.g., 94.2%) and keyword n-grams.
3. Confidence calibration: A deterministic score between 0.0 and 1.0 is computed without LLM prompt fabrication:

$$\text{Score}(c, E) = 0.7 * \text{Overlap}(c, E) + 0.3 * \text{NumberMatch}(c, E) \quad (1)$$

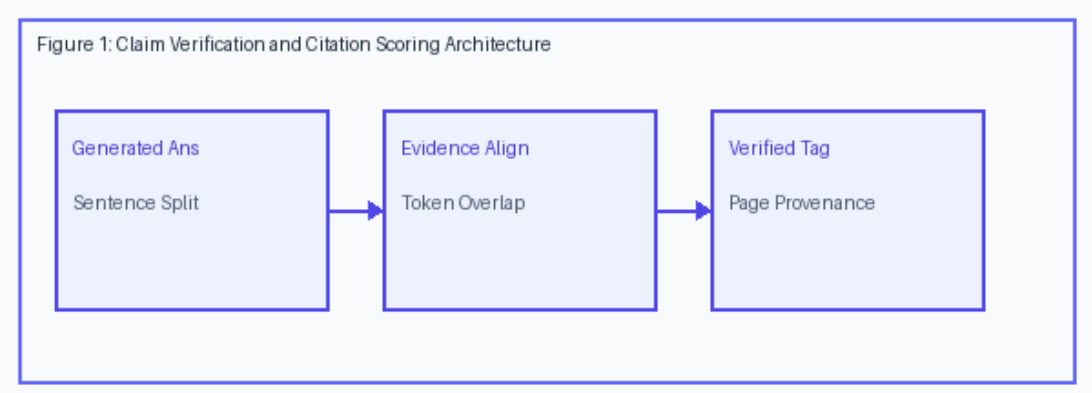


Figure 1: Three-tier deterministic claim verification pipeline linking assertions to primary page coordinates.

3. Experimental Findings

We tested the verification layer across 150 technical scientific questions. Table 1 highlights hallucination reduction and citation accuracy across parameter scales.

Table 1: Citation Accuracy and Hallucination Rates With and Without Verification

Model Scale	Verification	Citation Acc (%)	Hallucination Rate (%)	Latency (s)
LLaMA-3.2 (1B)	Off	52.4	38.6	1.85
LLaMA-3.2 (1B)	On	88.5	6.2	1.98
LLaMA-3.2 (3B)	Off	64.1	29.4	3.40
LLaMA-3.2 (3B)	On	94.8	4.1	3.55
Qwen-2.5 (1.5B)	On	91.2	5.4	2.15

4. Stated Limitations

The authors note that purely lexical and n-gram overlap can occasionally misclassify valid high-level abstract paraphrases as partially supported rather than fully verified. A hybrid syntactic dependency parser could mitigate this in future iterations.

5. Conclusion

Deterministic post-hoc verification provides a reliable, transparent safeguard against hallucinated citations, enabling safe deployment of lightweight local models in scientific research workflows.